

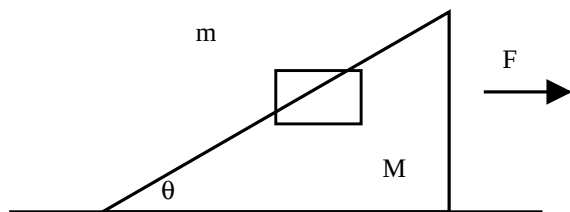
F=ma Practice - Day: Setup & Constraints

Goal: identify the correct physical framework before doing algebra. This set targets moving constraints, coupled motion, radial conditions, and multi-stage conservation problems.

Time target	45-50 min
Questions	12 multiple choice
Calculator	Allowed
Reference sheet	Do not use one
Strategy	For each question, spend the first 15-30 s deciding the governing idea before writing equations. If stuck for >4 min, m

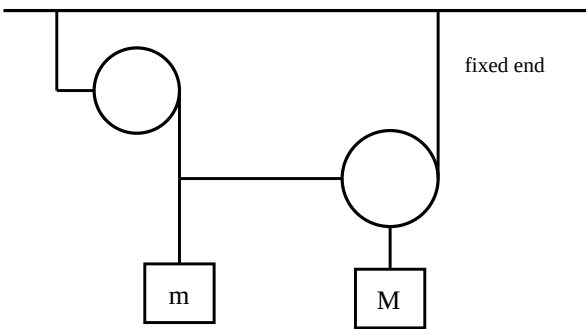
Important: Write enough work that I can see your setup choices when you send this back. I care more about diagnosing the setup than about arithmetic mistakes.

1. A block of mass m sits on a frictionless wedge of mass M and angle θ . The wedge is on a frictionless floor. A horizontal force F is applied to the wedge so that the block remains at a fixed point on the wedge. What must F be?



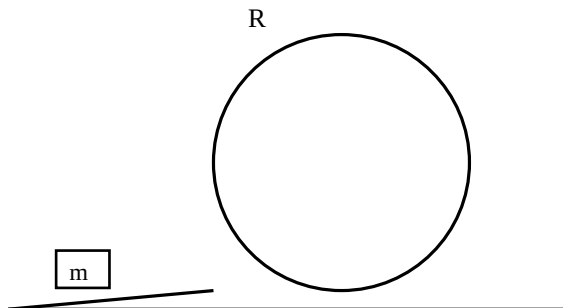
- (A) $(M+m)g \tan \theta$
- (B) $(M+m)g \cot \theta$
- (C) $Mg \tan \theta$
- (D) $mg \tan \theta$
- (E) $(M+m)g \sin \theta$

2. In the pulley system shown, all ropes and pulleys are massless and frictionless. The upper rope has fixed total length. Let downward be positive for both masses. If the moving pulley (and mass M attached to it) has acceleration a_M , what is the acceleration a_m of mass m ?



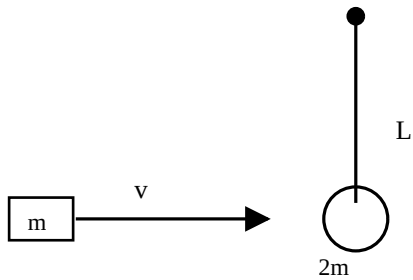
- (A) $a_m = -2a_M$
- (B) $a_m = -a_M/2$
- (C) $a_m = a_M$
- (D) $a_m = 2a_M$
- (E) $a_m = -a_M$

3. A small block starts from rest at height h on a frictionless track and enters a vertical circular loop of radius R . The block is not attached to the track. What is the minimum h , measured from the bottom of the loop, for the block to remain in contact all the way around?



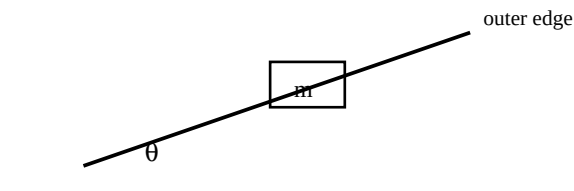
- (A) $2R$
- (B) $5R/2$
- (C) $3R$
- (D) $7R/2$
- (E) $4R$

4. A projectile of mass m moving horizontally at speed v embeds in a stationary pendulum bob of mass $2m$ and length L . The combined mass then swings upward. What is the minimum v required for the pendulum to complete a full vertical circle with the string taut?



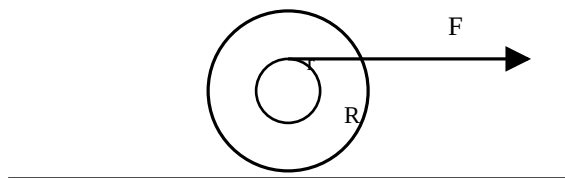
- (A) $3\sqrt{gL}$
- (B) $3\sqrt{2gL}$
- (C) $3\sqrt{5gL}$
- (D) $\sqrt{15gL}$
- (E) $2\sqrt{5gL}$

5. A car rounds a banked circular curve of radius R at speed v . The road is banked at angle θ , with the outer edge higher. There is no friction. Which relation must hold?



- (A) $\tan \theta = gR/v^2$
- (B) $\tan \theta = v^2/(gR)$
- (C) $\sin \theta = v^2/(gR)$
- (D) $\cos \theta = v^2/(gR)$
- (E) $\tan \theta = v/(gR)$

6. A spool of mass M has outer radius R and inner hub radius r . A string is wound around the top of the inner hub and pulled horizontally to the right with force F . The spool rolls without slipping on a horizontal surface. Its moment of inertia about its center is I . Which pair of equations is the correct starting point? Take rightward translation and clockwise rotation as positive.



- (A) $F+f=Ma$ and $Fr+fR=I\alpha$, with $a=\alpha R$
- (B) $F+f=Ma$ and $Fr-fR=I\alpha$, with $a=\alpha R$
- (C) $F-f=Ma$ and $Fr+fR=I\alpha$, with $a=\alpha r$
- (D) $F-f=Ma$ and $FR-fr=I\alpha$, with $a=\alpha R$
- (E) $F+f=Ma$ and $FR-fr=I\alpha$, with $a=\alpha r$

7. Two blocks of masses m and $2m$ lie on a frictionless horizontal table and are connected by a massless spring. The spring is initially compressed and the system is held at rest. After release, when the spring first reaches its natural length, the lighter block has speed u . What is the speed of the heavier block at that instant?

- (A) $u/4$
- (B) $u/2$
- (C) u
- (D) $2u$
- (E) $4u$

8. A small bead slides frictionlessly on a circular hoop of radius R that can move horizontally without friction. The hoop has mass M and the bead has mass m . Initially the hoop is at rest and the bead is released from rest at the top. Which quantity is conserved for the bead+hoop system during the motion?

- (A) The bead's mechanical energy only
- (B) Horizontal momentum only
- (C) Mechanical energy only
- (D) Both horizontal momentum and mechanical energy
- (E) Linear momentum in both directions

9. A block rests on a rough incline of angle θ . The entire incline accelerates horizontally to the right with acceleration A . In the non-inertial frame of the incline, which statement best determines the direction of static friction on the block?

- (A) Friction must always point up the incline.
- (B) Friction must always point down the incline.
- (C) Friction opposes the component, along the incline, of gravity plus the fictitious force.
- (D) Friction opposes the block's velocity relative to the ground.
- (E) Friction is zero whenever $A=g$.

10. A mass m hangs from a string of length L and moves as a conical pendulum, making angle θ with the vertical. Which single relation is sufficient to determine the angular speed ω ?

- (A) $T=mg$
- (B) $T \sin\theta = mg$
- (C) $T \cos\theta = m\omega^2 L$
- (D) $\omega^2 = g/(L \cos\theta)$
- (E) $\omega^2 = g \cos\theta/L$

11. A block of mass m slides down a rough incline of angle θ with coefficient of kinetic friction μ . It starts from rest and moves distance s along the incline. Which expression gives its speed after traveling s ?

- (A) $\sqrt{[2gs(\sin\theta - \mu\cos\theta)]}$
- (B) $\sqrt{[2gs(\cos\theta - \mu\sin\theta)]}$
- (C) $\sqrt{[2gs(\sin\theta + \mu\cos\theta)]}$
- (D) $\sqrt{[gs(\sin\theta - \mu\cos\theta)]}$
- (E) $\sqrt{[2gs(\mu\sin\theta - \cos\theta)]}$

12. A uniform rod of length L and mass M is initially vertical and hinged frictionlessly at its lower end. It is given an infinitesimal push and falls. What is the speed of its upper end just before the rod becomes horizontal?

- (A) $\sqrt{(gL)}$
- (B) $\sqrt{(3gL/2)}$
- (C) $\sqrt{(2gL)}$
- (D) $\sqrt{(3gL)}$
- (E) $2\sqrt{(gL)}$

Answer Sheet + Setup Notes

Record your answer and, in 3-8 words, the main framework you chose (for example: "string-length constraint", "momentum then energy", "radial force at top").

Q	Ans	Main setup / framework	Confidence (H/M/L)
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			

When you send this back: include your written work. I will grade both correctness and whether you selected the right model/constraint quickly.